

01-16B EMISSION SYSTEM [WL-C, WE-C]

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EMISSION SYSTEM OUTLINE [WL-C, WE-C]

dcf011600000t04

Feature

Improved exhaust gas purification	• EGR system adopted • EGR cooler adopted • Oxidation catalytic converter adopted
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Specification

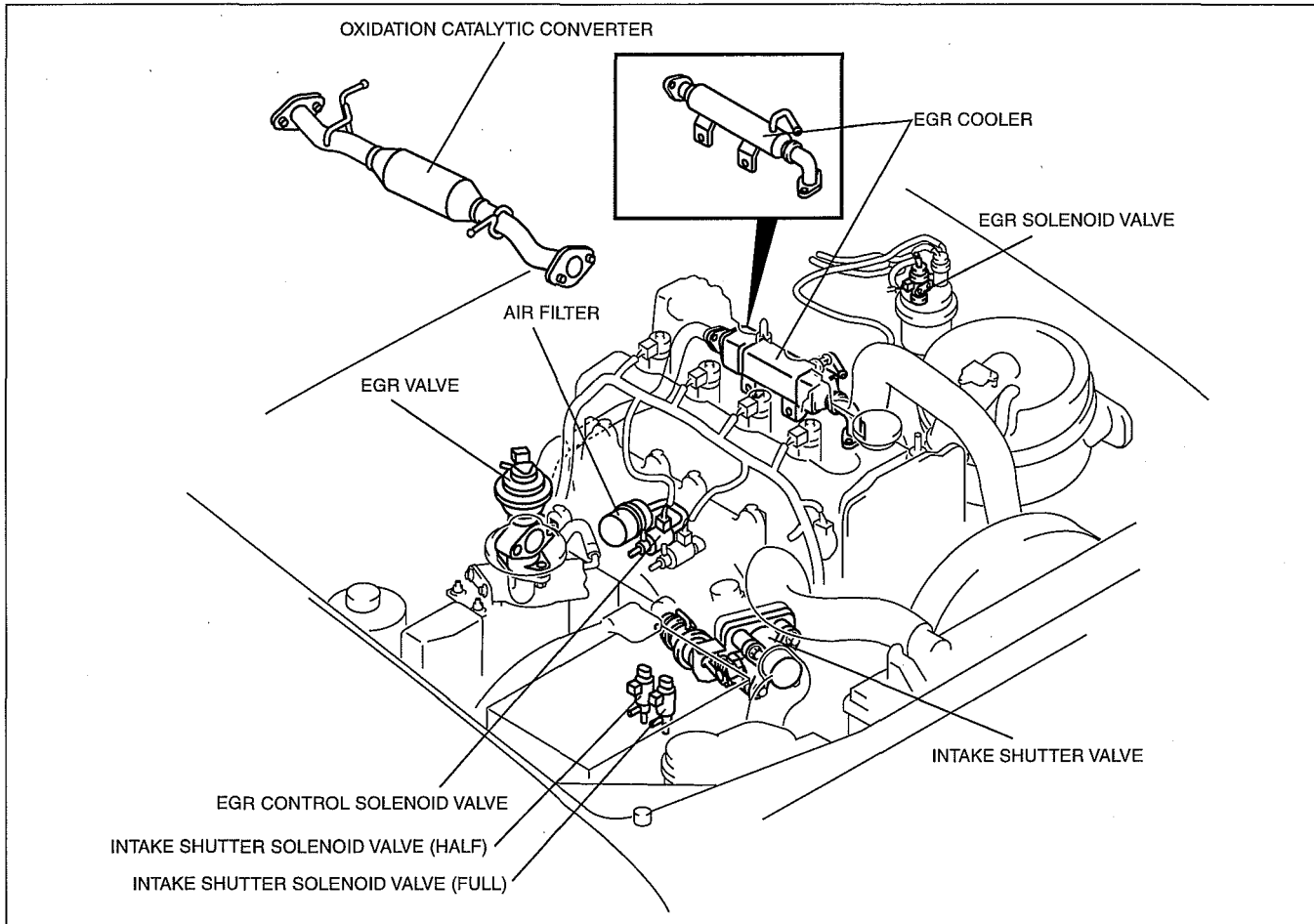
Item	WL-C, WE-C
EGR valve type	Vacuum
Catalytic converter type	Oxidation catalytic converter

EMISSION SYSTEM [WL-C, WE-C]

EMISSION SYSTEM STRUCTURAL VIEW [WL-C, WE-C]

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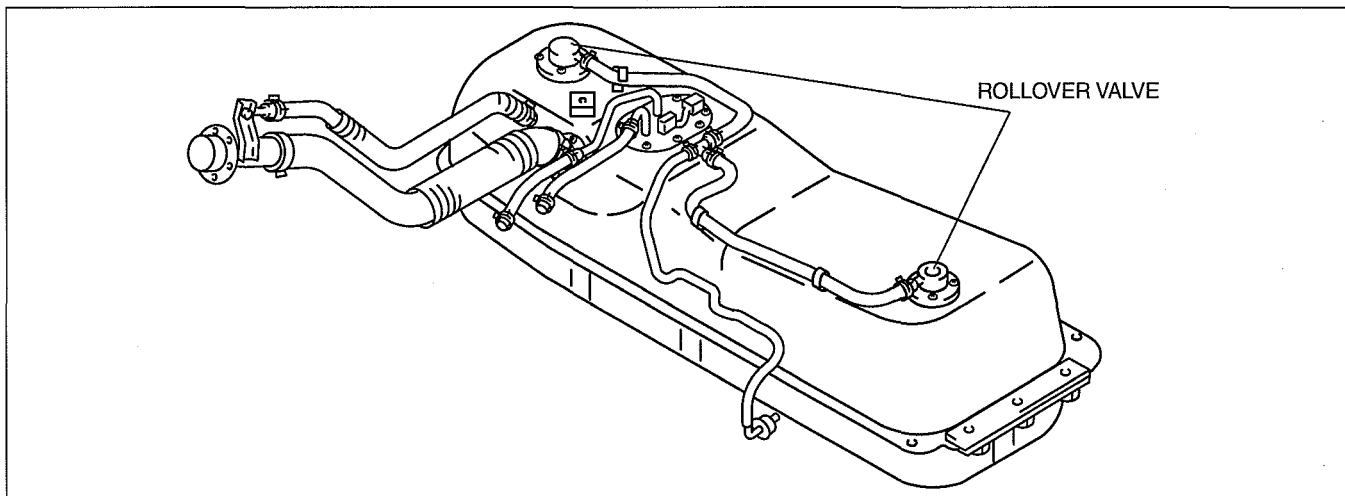
Engine Room Side



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Fuel Tank Side

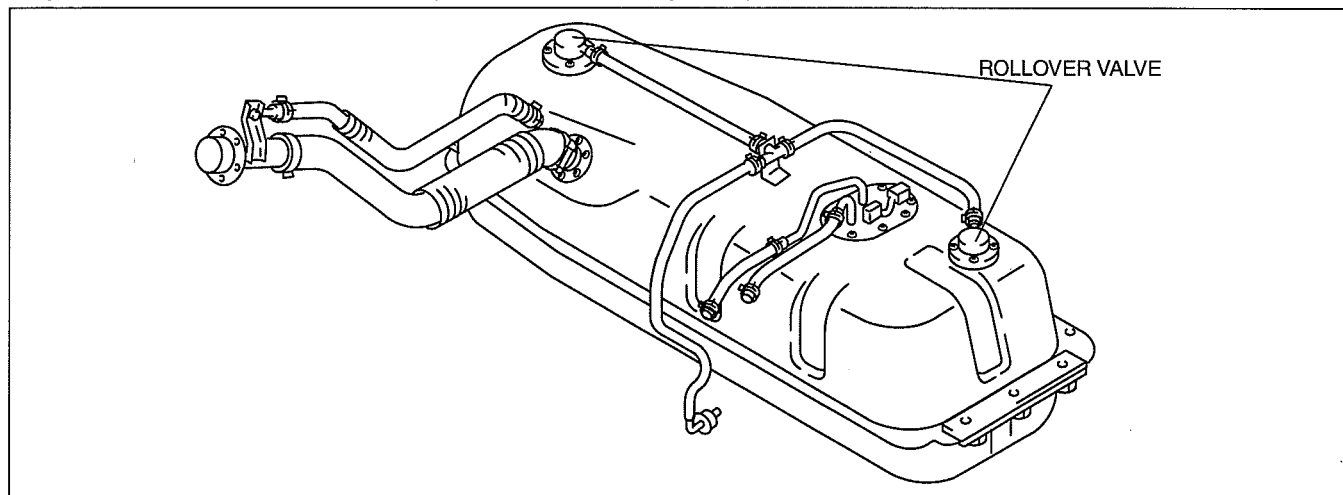
Double cab 4x2 (except Hi-Rider), Stretch cab 4x2 (with rear access system (except Hi-Rider))



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EMISSION SYSTEM [WL-C, WE-C]

Regular cab 4x4, stretch cab 4x4 (with rear access system), Hi-Rider

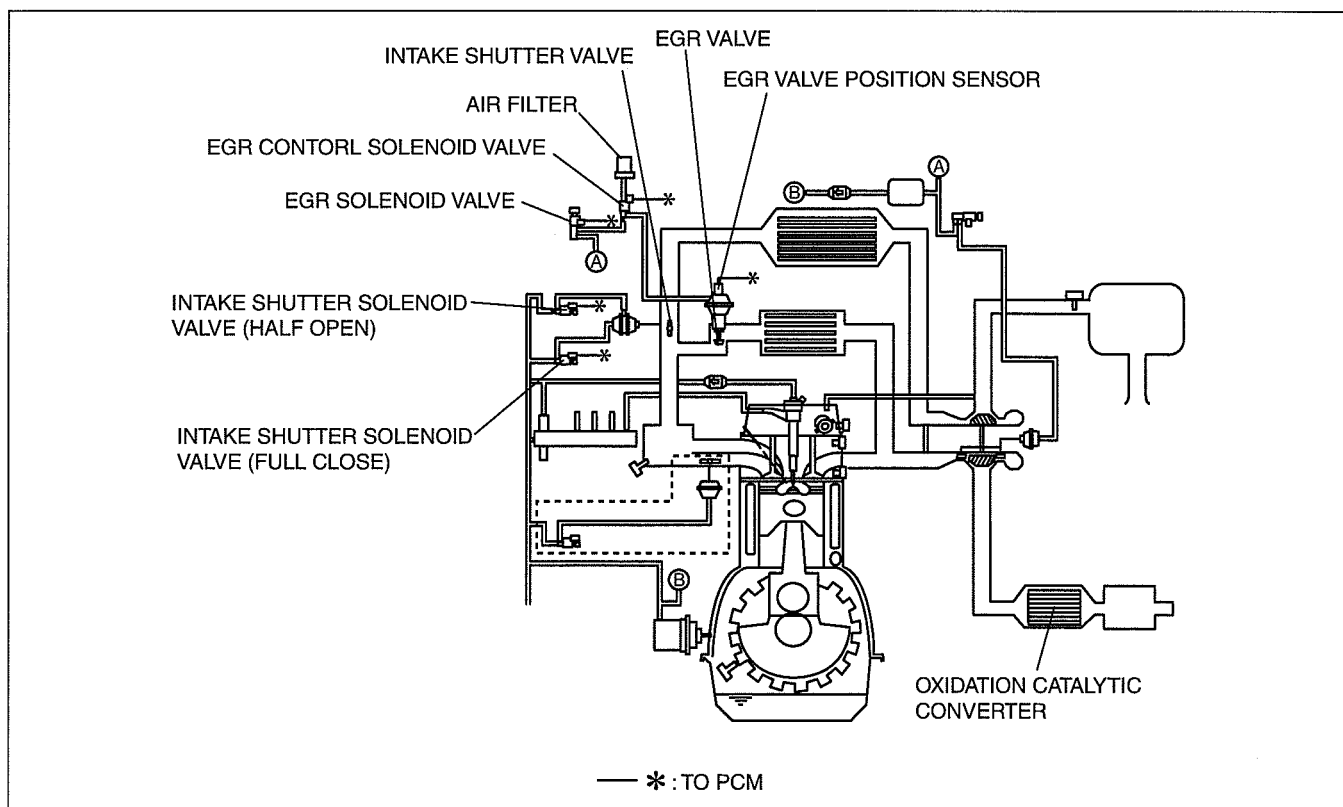


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EMISSION SYSTEM DIAGRAM [WL-C, WE-C]

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EMISSION SYSTEM [WL-C, WE-C]

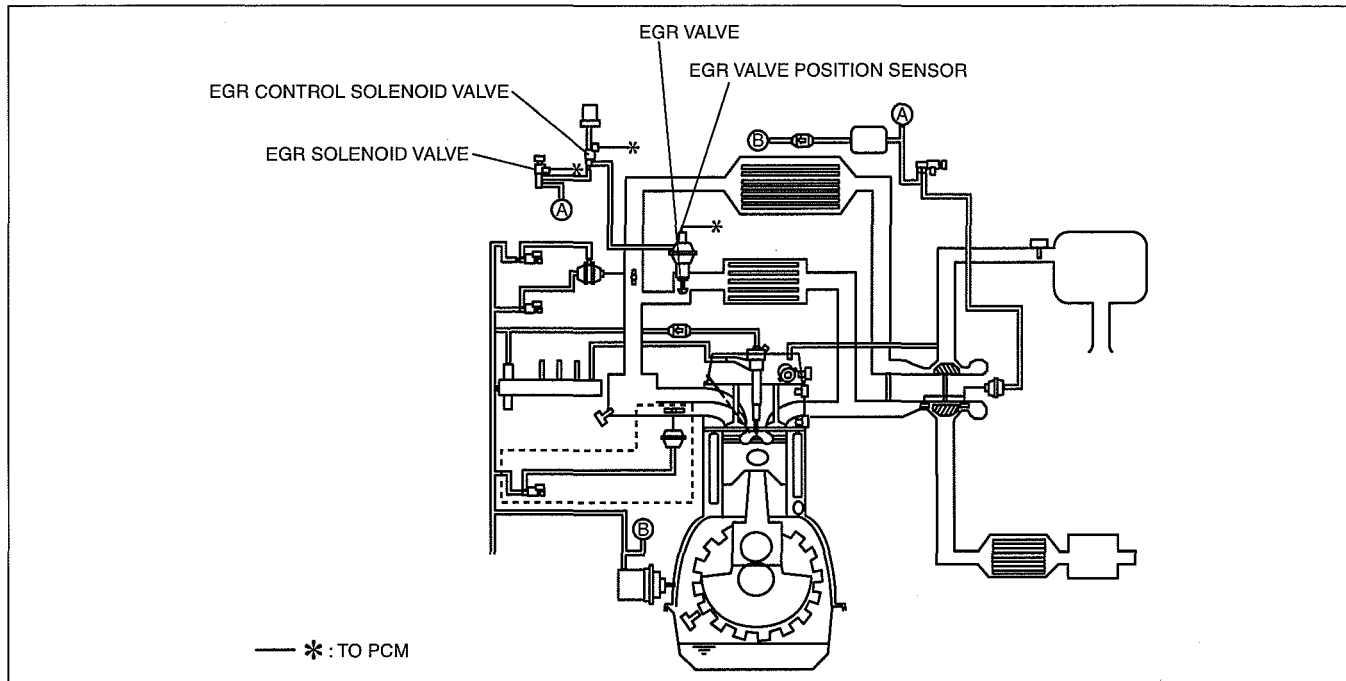
EGR SYSTEM OUTLINE [WL-C, WE-C]

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- The EGR cooler cools EGR gas and improves the intake-air charging efficiency, reducing black smoke.
- For control of EGR system, refer to CONTROL SYSTEM, EGR CONTROL. (See 01-40B-17 EGR CONTROL OUTLINE [WL-C, WE-C].) (See 01-40B-17 EGR CONTROL BLOCK DIAGRAM [WL-C, WE-C].) (See 01-40B-17 EGR CONTROL OPERATION [WL-C, WE-C].)

EGR SYSTEM DIAGRAM [WL-C, WE-C]

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EGR SYSTEM OPERATION [WL-C, WE-C]

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- To open the EGR valve and increase the EGR amount, increase the EGR solenoid valve control duty valve to increase the vacuum applied to the EGR valve. To close the EGR valve to decrease the EGR amount, increase the EGR control solenoid valve control duty valve to decrease the vacuum applied to the EGR valve.
- When the PCM energizes the intake shutter solenoid valve during the exhaust gas introduction at low engine speed, the vacuum passage is opened and vacuum is applied to the intake shutter valve actuator. Due to this, the intake shutter valve is closed and vacuum is generated downstream of the intake manifold after the intake shutter valve, realizing easier introduction of the exhaust gas to the combustion chamber.

EGR SYSTEM FUNCTION [WL-C, WE-C]

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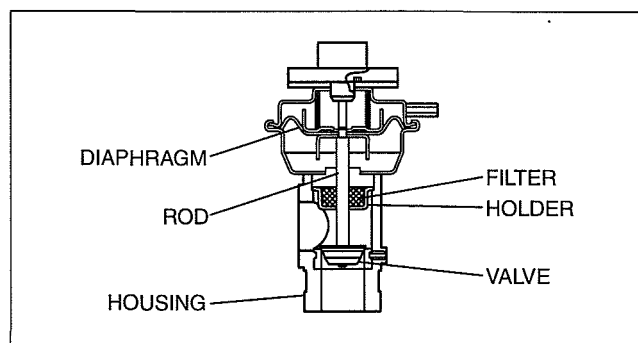
- The EGR valve, installed to the intake manifold, recirculates the exhaust gas to the intake manifold.

EMISSION SYSTEM [WL-C, WE-C]

EGR SYSTEM OPERATION [WL-C, WE-C]

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- Consists of a diaphragm and valves.
- The EGR solenoid valve operates based on the signal from the PCM and changes the valve opening angle to open the passage leading from the exhaust gas to the intake manifold by inducing the vacuum to the diaphragm on the upper part of the EGR valve.



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EGR SOLENOID VALVE FUNCTION [WL-C, WE-C]

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- EGR solenoid valve open/close the passage which brings vacuum generated in the vacuum pump to the diaphragm, and control the EGR valve opening.

EGR SOLENOID VALVE DESCRIPTION [WL-C, WE-C]

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Structure/Operation

- Vent side port of solenoid valve is connected to the EGR valve diaphragm through EGR control solenoid valve, and the vacuum side port of solenoid valve is connected to vacuum pump.
- Solenoid valve consists of a solenoid coil, spring, plunger and air filter.
- EGR solenoid valve increases the angle of the valve opening by increasing vacuum on EGR valve via the EGR control signal (duty signal).
- EGR solenoid valve decreases the angle of the valve opening by releasing vacuum on EGR valve via the EGR control signal (duty signal).

EGR CONTROL SOLENOID VALVE FUNCTION [WL-C, WE-C]

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- EGR valve changes the pressure on the diaphragm of the EGR valve from vacuum to atmospheric pressure.

EGR CONTROL SOLENOID VALVE DESCRIPTION [WL-C, WE-C]

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Structure/Operation

- EGR control solenoid valve consists of coil, spring, and plunger.
- When EGR valve is not energized, the passage between the diaphragm and atmospheric air side is open.
- When specified conditions are satisfied, the solenoid coils become energized and electromagnetic, and then pull the plunger. At this time, the passage between the diaphragm and atmospheric air side is closed and the passage between the diaphragm and vacuum pump side is opened.

EGR COOLER FUNCTION [WL-C, WE-C]

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- EGR cooler cools recirculated exhaust gas from the intake manifold.

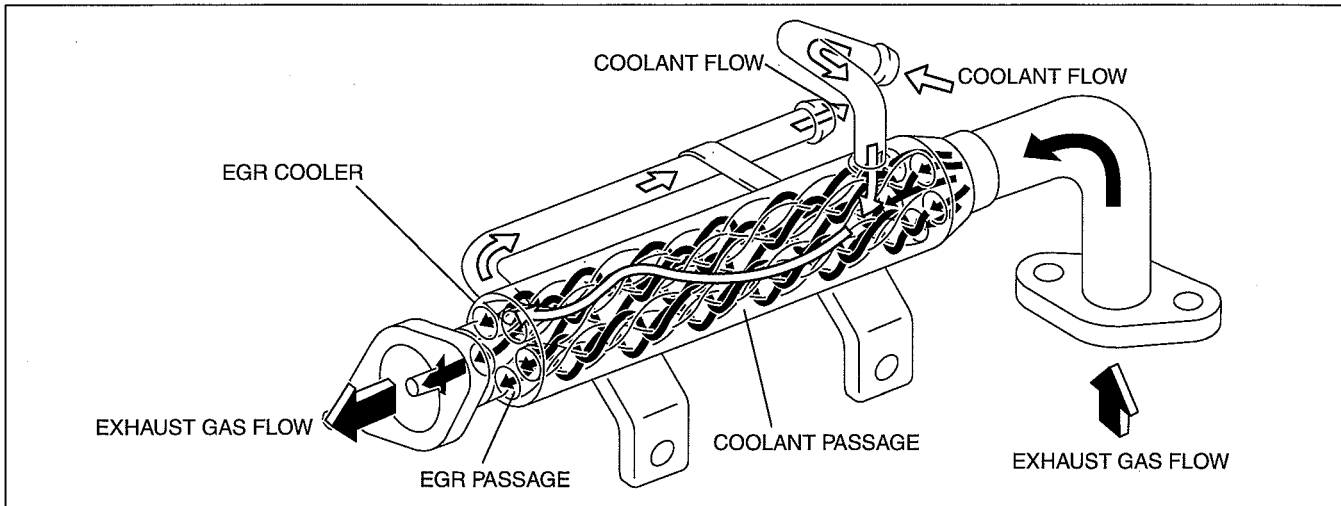
EMISSION SYSTEM [WL-C, WE-C]

EGR COOLER DESCRIPTION [WL-C, WE-C]

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Structure/Operation

- EGR cooler has an EGR passage and coolant passage.



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INTAKE SHUTTER VALVE DESCRIPTION [WL-C, WE-C]

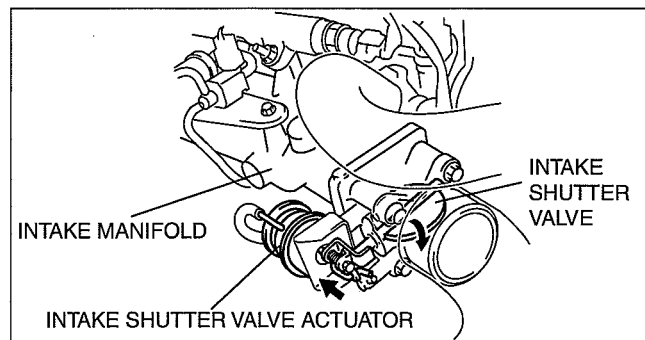
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Function

- When the exhaust gas recirculation system is operated, the intake shutter valve closes, vacuum is generated in the intake manifold, and EGR gas is efficiently transported to the combustion chamber. As a result, NOx is reduced.
- When the engine is idling, the intake shutter valve closes and reduces air intake noise.

Structure/Operation

- When the exhaust gas recirculation system is operated and vacuum is applied to the diaphragm of the intake shutter valve actuator, internal lever is pulled and the intake shutter valve closes.



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INTAKE SHUTTER SOLENOID VALVE DESCRIPTION [WL-C, WE-C]

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Function

- The intake shutter solenoid valve switches the opening/closing of the passage where the vacuum generated in the vacuum pump is led to the diaphragm of the intake shutter valve actuator.

Structure

- The intake shutter valve consists mainly of a solenoid coil, and a spring, a plunger.

Operation

- The vacuum passage between the intake shutter valve actuator and the vacuum pump is opened or closed depending upon whether the intake shutter solenoid valve is energized or de-energized.

OXIDATION CATALYTIC CONVERTER FUNCTION [WL-C, WE-C]

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- Purifies contaminants in the exhaust gas by utilizing a chemical reaction in a three way catalytic converter.
- The catalytic converter converts the noxious carbon monoxide and hydrocarbon in the exhaust gas to non-noxious carbon dioxide and water.

EMISSION SYSTEM [WL-C, WE-C]

OXIDATION CATALYTIC CONVERTER CONSTRUCTION [WL-C, WE-C]

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- Consists of a three-way catalytic converter and insulator.
- A catalytic converter utilizing a platinum–palladium–rhodium system has been adopted.

OXIDATION CATALYTIC CONVERTER OPERATION [WL-C, WE-C]

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- Contaminants in the exhaust gas (HC and CO) are purified by oxidization while passing through the catalytic converter.
 - Oxidization process
 - Noxious hydrocarbon (HC) and carbon monoxide (CO) are bonded to oxygen which is converted to non-noxious carbon dioxide and water.
$$\text{O}_2 + \text{HC} + \text{CO} \rightarrow \text{CO}_2 + \text{H}_2\text{O}$$

ROLLOVER VALVE FUNCTION [WL-C, WE-C]

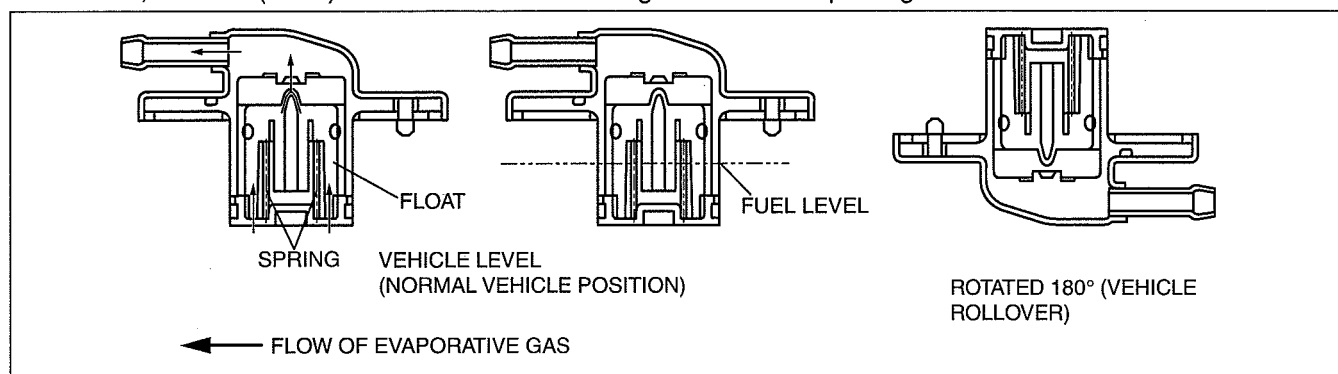
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- The rollover valve prevents fuel flow into the airflow hose during sudden cornering or vehicle rollover.

ROLLOVER VALVE CONSTRUCTION/OPERATION [WL-C, WE-C]

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- The rollover valve consists of a float, and spring.
- The rollover valve utilizes a combination of float weight, spring force, and buoyancy. When the float is sunk in the fuel, the float (valve) closes to block the sealing surface of the passage.



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